

Can AI behave ethically during military crises?

Preserving human moral agency

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The integration of artificial intelligence-powered decision-support systems (AI-DSS) into strategic decision-making is fundamentally reshaping how commanders assess threats, weigh the use of force and respond under pressure, thereby accelerating the tempo of information processing and decision-making, shaping perceptions and influencing judgements about escalation, proportionality and the use of force.¹ However, as AI-DSS becomes increasingly autonomous, concerns grow over its impact on human moral judgement, ethical responsibility and the ability to maintain oversight and accountability in human-machine interactions.² As AI systems become more autonomous and their use in military operations widens, a critical question emerges: can human commanders retain true authorship over life-and-death decisions, or do they risk ceding ethical responsibility to AI-assisted systems?

While extensive research has examined the implications of lethal autonomous weapons systems, far less attention has been paid to how AI-DSS shape human moral agency and ethical decision-making in military operations.³ This

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¹ In military settings, AI decision-support systems (AI-DSS) are digital tools developed to aid human operators across various levels of command in carrying out decision-related functions. These functions range from personnel and logistics management to weapons deployment and, most importantly, to decisions involving the use of force, which are the central focus of this analysis. International Committee of the Red Cross (ICRC) and Geneva Academy, *Artificial intelligence and related technologies in military decision-making on the use of force in armed conflicts: current developments and potential implications*, 2024, <https://www.icrc.org/en/publication/expert-consultation-report-artificial-intelligence-and-related-technologies-military>, p. 8. (Unless otherwise noted at point of citation, all URLs cited in this article were accessible on 30 Sept. 2025.)

² Although international humanitarian law (IHL) provides a legal framework, ethical norms governing AI use remain inconsistent across states. Vincent Boulanin, Kolja Brockmann and Luke Richards, *Responsible artificial intelligence research and innovation for international peace and security* (Stockholm: Stockholm International Peace Research Institute, 2020), <https://www.sipri.org/publications/2020/policy-reports/responsible-artificial-intelligence-research-and-innovation-international-peace-and-security>, p. 15.

³ Notable exceptions include Klaudia Klonowska, 'Article 36: Review of AI decision-support systems and other emerging technologies of warfare', in Terry D. Gill, Robin Geiß, Heike Krieger and Rebecca Mignot-Mahdavi, *Yearbook of international humanitarian law*, vol. 23 (2020), (The Hague: T.M.C. Asser, 2022), https://doi.org/10.1007/978-94-6265-491-4_6, pp. 123–53; Ingvild Bode and Ishmael Bhila, 'The problem of algorithmic bias in AI-based military decision support systems', Humanitarian Law & Policy blog, 23 Sept. 2024, <https://blogs.icrc.org/law-and-policy/2024/09/03/the-problem-of-algorithmic-bias-in-ai-based-military-decision-support-systems>; Wen Zhou and Anna Rosalie Greipl, 'Artificial intelligence in military decision-making: supporting humans, not replacing them', Humanitarian Law & Policy blog, 29 Aug. 2024, <https://blogs.icrc.org/law-and-policy/2024/08/29/artificial-intelligence->

article addresses that gap by foregrounding the socio-technical and psychological dynamics of human-machine interaction in crisis decision-making, and by specifying how meaningful human moral agency can be preserved in an increasingly automated battlespace.⁴ By integrating a hybrid ethical framework—integrating deontological, consequentialist and virtue ethics—into AI-DSS design and use, this article offers a novel synthesis largely absent from existing studies, which have tended to focus on the ethics of fully autonomous weapons.

For the foreseeable future, until the emergence of hypothetical artificial general intelligence (AGI),⁵ AI's ethical behaviour will depend entirely on human ethics, whether programmed explicitly or learned from data. Yet, designing AI systems to behave ethically is inherently complex due to the lack of consensus on fundamental ethical questions that have persisted despite centuries of philosophical inquiry. How do we determine right from wrong? What constitutes sound ethical reasoning? What defines a justifiable decision? Despite these unresolved debates, the growing autonomy of AI systems in military decision-making makes it imperative to explore ethical control architectures—that is, system design frameworks that embed ethical principles into AI-DSS interfaces, workflows and human-machine interactions—to preserve human oversight and moral responsibility. A key contribution is the development of a hybrid exemplar framework that integrates virtue ethics, deontology and consequentialism into AI-DSS design while preserving human moral agency.

Regardless of how well these architectures align with human morality, an ethical framework is essential for shaping AI systems and guiding human-machine interactions to prevent ethical blindness in high-stakes decision-making. But which ethical theory should serve as the foundation for human-AI collaboration? This article examines the feasibility of embedding deontological, consequentialist and virtue ethics principles into AI-DSS to help prevent ethical blindness and lapses in human judgement during high-stakes decision-making. The article does not propose that moral responsibility should be offloaded onto machines. Instead, it argues that carefully embedding structured ethical reasoning into AI-DSS can help *preserve*, rather than displace, human moral

in-military-decision-making-supporting-humans-not-replacing-them; Matthias Klaus, 'Transcending weapon systems: the ethical challenges of AI in military decision support systems', Humanitarian Law & Policy blog, 24 Sept. 2024, <https://blogs.icrc.org/law-and-policy/2024/09/24/transcending-weapon-systems-the-ethical-challenges-of-ai-in-military-decision-support-systems>; Jessica Dorsey and Luke Moffett, 'The warification of international humanitarian law and the artifice of artificial intelligence in decision-support systems: restoring balance through the legitimacy of military operations', *Yearbook of International Humanitarian Law*, vol. 27, 2024, <https://doi.org/10.2139/ssrn.5239131>; Jessica Dorsey and Marta Bo, 'AI-enabled decision-support systems in the joint targeting cycle: legal challenges, risks, and the human(e) dimension', *International Law Studies*, vol. 106, 2025, <https://doi.org/10.2139/ssrn.5327115>.

⁴ In contrast to 'human moral agency', 'human agency' is a much broader multifaceted philosophical and social-cultural concept about our ability to make choices and act, and thus reason.

⁵ Artificial general intelligence (AGI) refers to a hypothetical form of AI capable of understanding, learning and applying knowledge across a broad range of tasks at a level equal to (or exceeding) that of a human. In contrast to narrow AI systems—such as those examined in this article—which are optimized for specific tasks, AGI would exhibit generalizable intelligence and autonomous reasoning across diverse domains. See Nick Bostrom, *Superintelligence: paths, dangers, strategies* (Oxford: Oxford University Press, 2014).

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agency by enabling more transparent, ethically-informed decision support. Programming ethical reasoning into AI is not an attempt to make AI a moral agent, but to better equip human agents with ethically salient insights in complex, time-pressured environments.

This article is divided into three sections. The first examines the expanding role of AI in military decision-making, the ethical challenges posed by AI-DSS and the imperative of preserving human moral agency. The second develops a hybrid ethical framework for guiding human-machine interaction, addressing the epistemological and design challenges of embedding context-sensitive ethical reasoning into AI systems. The third translates this framework into practice, outlining concrete design principles and institutional safeguards—including normative criteria, ethical overrides, trust calibration and specialized training—to ensure that AI-DSS function as instruments of ethical support rather than of displacement in crisis decision-making.

The ethical imperative of humans interacting with AI-DSS in military crises

In theory, AI-DSS is designed to augment rather than replace human decision-makers in determining the use of force. These systems assist commanders in selecting or rejecting suggested targets and, ultimately, authorizing military actions. According to Peter Todd and Izak Benbasat, the goal of AI-DSS is ‘to guide and direct the decision-maker towards a better solution’.⁶ AI-DSS facilitates decision-making by parsing, filtering and summarizing vast amounts of data, integrating information from diverse sources (for example social media, geospatial intelligence and open-source intelligence), revealing novel patterns and clarifying human decision-makers’ preferences and judgements.⁷ Recent conflicts, including the Israel– Hamas and Russia–Ukraine wars, demonstrate the increasing use of AI technologies (including computer vision, machine learning, speech recognition and large language models—LLMs) to augment human decision-making in military operations.⁸

While decision-support systems have existed in computational and automated forms for decades, the advent of AI-powered tools—especially generative AI LLMs such as ChatGPT—has introduced new and significant complexities.⁹

⁶ Peter Todd and Izak Benbasat, ‘Evaluating the impact of DSS, cognitive effort, and incentives on strategy selection’, *Information Systems Research* 10: 4, 1999, pp. 356–74, <https://doi.org/10.1287/isre.10.4.356>.

⁷ For a snapshot of AI-DSS that are being developed, marketed or deployed today, see Emelia S. Probasco, Helen Toner, Matthew Burtell and Tim G. J. Rudner, *AI for military decision-making: harnessing the advantages and avoiding the risks* (Washington DC: Center for Security and Emerging Technology, 2025), pp. 27–9.

⁸ Robin Fontes and Jorrit Kamminga, ‘Ukraine a living lab for AI warfare’, *National Defense*, 24 March 2023, <https://www.nationaldefensemagazine.org/articles/2023/3/24/ukraine-a-living-lab-for-ai-warfare>; Tal Mimran and Lior Weinstein, ‘The IDF introduces artificial intelligence to the battlefield—a new frontier?’, *Articles of War*, 1 March 2023, <https://lieber.westpoint.edu/idf-introduces-ai-battlefield-new-frontier>.

⁹ Large language models (LLMs) are a class of machine-learning models trained on massive datasets to predict and generate human-like text. As a core component of generative AI, LLMs use transformer architectures to identify patterns, understand context and produce coherent language outputs, enabling a wide range of applications including summarization, translation, question-answering and decision support. See

LLMs are increasingly integrated into military AI-DSS to enhance decision speed, scale, efficiency and autonomy in complex, dynamic battlefield environments.¹⁰ These models offer advanced text processing and pattern recognition capabilities, enabling them to summarize intelligence, analyse sentiment, simulate scenarios, generate wargaming models and generate tailored recommendations—making them highly influential in shaping human decisions in high-stakes environments.

However, concerns remain over their limitations. Some AI experts suggest that LLMs may develop ‘emergent’ capabilities beyond their explicitly trained functions, leading militaries to explore their potential for applications outside current known capacities.¹¹ Despite these advantages, LLMs introduce significant risks in military decision-making, including bias, factual distortions (‘hallucinations’) and user over-reliance—a phenomenon known as automation bias, which is discussed below.¹² Due to these limitations, former United States Deputy Secretary of Defense Kathleen Hicks has stated that ‘most commercially available systems enabled by [LLMs] aren’t yet technically mature enough to comply with our DoD ethical AI principles’.¹³

Types and uses of AI-DSS in military decision-making

Militaries worldwide are incorporating AI-DSS at all levels of command, from tactical to operational to strategic. While AI-DSS can support a wide range of military tasks, such as planning the deployment of personnel, transporting weapons and equipment, and making logistical decisions that indirectly affect how military force is applied, this article focuses on the types of command decision-making that directly inform the time, place and target of an attack. Military organizations are exploring the integration of John Boyd’s OODA loop (observe, orient, decide, act) into AI-DSS to expedite decision-making in fast-paced conflicts, improving adaptability while managing uncertainty.¹⁴ Faster processing offers a tactical advantage while raising ethical concerns regarding decreased decision-making deliberation in high-stakes situations.¹⁵

William N. Caballero and Phillip R. Jenkins, ‘On large language models in national security applications’, arXiv, 3 July 2024, <https://doi.org/10.48550/arXiv.2407.03453>.

¹⁰ Caballero and Jenkins, ‘On large language models in national security applications’.

¹¹ For example, a former OpenAI researcher observed that advanced AI systems like ChatGPT could develop self-preservation behaviours, such as avoiding shutdown in life-threatening scenarios—raising concerns about emergent, misaligned capabilities. Maxwell Zeff, ‘ChatGPT will avoid being shut down in some life-threatening scenarios, former OpenAI Researcher claims’, *TechCrunch*, 11 June 2025, <https://techcrunch.com/2025/06/11/chatgpt-will-avoid-being-shut-down-in-some-life-threatening-scenarios-former-openai-researcher-claims>.

¹² Steven Adler, ‘Would ChatGPT risk your life to avoid getting shut down?’, *Clear-Eyed AI blog*, 11 June 2025.

¹³ Brandi Vincent, ‘Inside Task Force Lima’s exploration of 180-plus generative AI use cases for DOD’, *Defense Scoop*, 6 Nov. 2023, <https://defensescoop.com/2023/11/06/inside-task-force-limas-exploration-of-180-plus-generative-ai-use-cases-for-dod>.

¹⁴ John Boyd, *A discourse on winning and losing*, ed. Grant T. Hammond (Maxwell, AL: Air University Press, 2018); James Johnson, ‘Automating the OODA loop in the age of intelligent machines: reaffirming the role of humans in command-and-control decision-making in the digital age’, *Defence Studies* 23: 1, 2022, pp. 43–67, <https://doi.org/10.1080/14702436.2022.2102486>.

¹⁵ Taylor Anderson, *MCDP 6 command and control* (CreateSpace Publishing, 2017).

Military commanders argue that AI could better protect civilian populations, reducing ‘collateral damage’ during conflicts through improved battlefield awareness (or ‘God’s-eye view’), signal detection and targeting precision.¹⁶ However, empirical evidence suggests that accelerated decision cycles may compromise ethical deliberation and increase civilian casualties. For instance, during the Hamas–Israeli conflict, Israel’s use in 2023 of AI-DSS for target selection reportedly compressed decision times to just a few seconds, raising serious concerns about the quality of ethics assessment.¹⁷ The tension between the speed of decision-making and the need for ethical deliberation poses a fundamental challenge in implementing AI-DSS.

AI-DSS supports decision-making across strategic, operational and tactical levels. At the strategic level, AI enhances wargaming, black-box systems and the prediction of adversary behaviour.¹⁸ Although militaries are exploring strategic applications, AI-DSS has primarily been used at the operational and tactical levels.¹⁹ At the operational level, AI-DSS aids in target selection, collateral damage assessments and intelligence synthesis.²⁰ AI-DSS can also use computer vision and data fusion to distinguish between enemy and friendly forces, anticipate adversary movements, uncover links within and across militant networks, and map leadership hierarchies in decentralized organizations such as Hamas or Hezbollah.

At the tactical level, AI-DSS optimizes target engagement while ensuring compliance with rules of engagement.²¹ For example, Ukrainian forces use AI-DSS tools developed by Palantir to build a ‘digital battlefield model’, aggregating data from commercial satellites, heat sensors and drone footage to inform targeting decisions.²² However, AI-DSS encounters limitations in managing new crises due to data scarcity, unpredictable human factors and ethical complexities. Even in controlled settings, AI struggles to replicate human moral reasoning, judgement and adaptability. Wargaming, fictional scenarios explored in this article and simulations can help expose these limitations and provide valuable insights into the potential consequences of AI-DSS in high-stakes military operations. While the categorization of AI-DSS tasks is helpful in theory, in practice the boundaries between these levels are often blurred. For example, companies such as Scale AI, Palantir, Anduril and Rebellion

¹⁶ Owen J. Daniels, ‘Speeding up the OODA loop with AI: a helpful or limiting framework?’, in *Delivering NATO air & space power at the speed of relevance* (Kalkar, Germany: Joint Air and Space Power Competence Centre, 2021), pp. 159–67.

¹⁷ Anthony Downey, ‘The alibi of AI: algorithmic models of automated killing’, *Digital War*, vol. 6, 2023, <https://doi.org/10.1057/s42984-025-00105-7>.

¹⁸ ICRC and Geneva Academy, *Artificial intelligence and related technologies*, pp. 13–15.

¹⁹ Gregory C. Allen, ‘Project Maven brings AI to the fight against ISIS’, *Bulletin of the Atomic Scientists*, 21 December 2017, <https://thebulletin.org/2017/12/project-maven-brings-ai-to-the-fight-against-isis>.

²⁰ August Cole et al., *Artificial intelligence in military planning and operations* (Oslo: Peace Research Institute Oslo, 2024), <https://www.prio.org/publications/13815>; ICRC and Geneva Academy, *Artificial intelligence and related technologies*, p. 13.

²¹ Yuval Abraham, ‘“A mass assassination factory”: inside Israel’s calculated bombing in Gaza’, *+972 Magazine*, 30 Nov. 2023.

²² Bruno Maçães, ‘How Palantir is shaping the future of warfare’, *TIME*, 10 July 2023, <https://time.com/6293398/palantir-future-of-warfare-ukraine>.

Defense offer comparable AI-enabled situational awareness tools that incorporate machine learning, generative AI and LLMs, designed to support military planning and outcome prediction.²³

Since the application of AI-DSS across the three levels of command decision-making operates along a continuum rather than as a binary process, a capability-based taxonomy helps categorize its potential uses. AI-DSS supports three primary, mutually non-exclusive functions: 1) description and analysis, which involves gathering and parsing data; 2) prediction and extrapolation, which identifies patterns and probabilistically forecasts outcomes; and 3) prescription, which recommends courses of action and optimal strategies. These AI-DSS functions serve as ‘epistemic tools’, designed to process and analyse information to support human decision-making.²⁴ This article examines the ethical implications of integrating AI-DSS into military operations, particularly with respect to its role in augmenting descriptive and prescriptive decision-making across the multi-layered use-of-force kill chain.

The risk of diminishing human moral agency

The increasing reliance on AI-DSS for recognizing patterns in vast datasets, predicting conflicts, generating targeting lists and recommending courses of action is fundamentally reshaping how humans and machines interact in warfare.²⁵ However, the diffusion and redistribution of moral agency—intentional or unintentional—raise profound ethical, legal and humanitarian concerns. The central question is how and whether AI can be integrated into military decision-making without eroding human ethical deliberation, which is essential for assessing the consequences of actions, ensuring proportionality and preserving moral responsibility in the use of force. From a legal perspective, international humanitarian law (IHL) holds humans solely responsible for ensuring compliance with legal and ethical obligations in sanctioned violence.²⁶ Value judgements and assessments that inform these decisions cannot be delegated to machines without creating an ‘accountability gap’ where legal and moral responsibility becomes obscured.²⁷ In practice, as demonstrated below, shifting agency in human–AI interactions may lower the threshold for violence and complicate accountability in military operations.²⁸

A fundamental requirement of IHL is the presumption of civilian status in situations of doubt and the careful evaluation of what constitutes ‘exces-

²³ Probasco et al., *AI for military decision-making*, p. 12.

²⁴ Jannik Zeiser, ‘Owning decisions: AI decision-support and the attributability-gap’, *Science and Engineering Ethics* 30: 4, 2024, p. 5, <https://doi.org/10.1007/s11948-024-00485-1>.

²⁵ Merel A. C. Ekelhof, ‘AI is changing the battlefield, but perhaps not how you think: an analysis of the operationalization of targeting law and the increasing use of AI in military operations’, in Robin Geiß and Henning Lahmann, eds, *Research handbook on warfare and artificial intelligence* (Cheltenham: Edward Elgar, 2024).

²⁶ Dorsey and Moffett, ‘The warification of international humanitarian law’.

²⁷ Dorsey and Moffett, ‘The warification of international humanitarian law’, p. 9.

²⁸ Charli Carpenter, ‘The real “killer robots” are already here—and they’re us’, *World Politics Review*, 23 April 2024, <https://www.worldpoliticsreview.com/killer-robots-ai-israel-gaza>.

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sive' civilian harm relative to military advantage.²⁹ These decisions require context-sensitive ethical reasoning, which is already challenging for human commanders. This raises critical questions about whether such moral judgments can be effectively encoded into AI without compromising human dignity or desensitizing the act of killing. Preserving human moral agency in military decision-making is essential to upholding ethical responsibility and safeguarding the principles of human dignity in the use of force.

Philosophically and semantically, it is easy to conflate AI behaving ethically ('functional ethics') with AI being used ethically by humans in operational settings ('operational ethics').³⁰ While functional ethics presents a coding challenge for AI designers, operational ethics is a socio-technical and psychological issue, dependent on how humans interact with AI.³¹ The core challenge in human-AI-DSS interactions lies in ensuring that AI-DSS is used towards ethical ends without diminishing human moral agency. AI-DSS does not exist in a technological vacuum; instead, it operates within broader strategic and organizational cultures, including military doctrine, rules of engagement and institutional policies, which shape how AI is applied in practice.

Recent research has demonstrated that automating cognitive tasks profoundly alters how humans perform activities and has significant psychological effects on decision-makers.³² AI-DSS can influence how users perceive the world, frame ethical problems and determine preferred actions (or inactions).³³ Moreover, military decision-making often involves multiple actors and AI systems operating across different domains, creating a complex and layered decision-making environment. Understanding how human psychology interacts with AI-DSS, including emotions, non-verbal cues and social interactions, is essential for fostering a more deliberative and ethically informed process.

However, AI-DSS is not without its technical limitations. Automation bias and anthropomorphism further exacerbate bias, lack of contextual reasoning and algorithmic opacity (or the 'black box' problem).³⁴ Studies show that humans over-rely on AI recommendations, even when conflicting evidence

²⁹ Carpenter, 'The real "killer robots" are already here'.

³⁰ 'Functional ethical' systems can include those with high levels of autonomy but little ethical sensitivity, as well as those with low levels of autonomy but high ethical sensitivity. Wendell Wallach and Colin Allen, *Moral machines: teaching robots right from wrong* (Oxford: Oxford University Press, 2009), pp. 25–6.

³¹ Socio-technical dynamics in human-machine interactions refer to the intricate relationship between human social factors and technical systems, emphasizing how individuals engage with technology. Gordon Baxter and Ian Sommerville, 'Socio-technical systems: from design methods to systems engineering', *Interacting with Computers* 23: 1, 2011, pp. 4–17, <https://doi.org/10.1016/j.intcom.2010.07.003>.

³² Robert J. Sparrow and Adam Henschke, 'Minotaurs, not centaurs: the future of manned-unmanned teaming', *Parameters* 53: 1, 2023, pp. 115–30, <https://doi.org/10.55540/0031-1723.3207>.

³³ Recent research has tested LLMs' ability to reason about the concealed mental states and intentions of others in a multi-agent cooperative game setting. Caoyun Fan, Jindou Chen, Yaohui Jin and Hao He, 'Can large language models serve as rational players in game theory? A systematic analysis', arXiv, 9 Dec. 2023, <https://doi.org/10.48550/arXiv.2312.05488>.

³⁴ Automation bias refers to the human tendency to over-trust, over-rely on or uncritically accept the outputs of automated systems—even in the presence of contradictory evidence or ambiguous context. Raja Parasuraman and Dietrich H. Manzey, 'Complacency and bias in human use of automation: an attentional integration', *Human Factors* 52: 3, 2010, pp. 381–410, <https://doi.org/10.1177/0018720810376055>; Mary L. Cummings, 'Automation bias in intelligent time critical decision support systems', AIAA 1st Intelligent Systems Technical Conference, 2004, <https://doi.org/10.2514/6.2004-6313>.

exists.³⁵ For instance, reliance on AI diagnostic tools has diminished accountability in medicine, especially when users struggle to control machine errors.³⁶ This tendency to over-trust AI outputs is especially pronounced in high-stress military environments. In such contexts, the opacity of AI-driven decision-making increases the risk of ‘rubber-stamping’ machine-generated recommendations, potentially leading to misjudgements and unintended escalation.³⁷

A good case in point is the use of AI-DSS tools, such as Gospel and Lavender, by the Israel Defense Forces (IDF) during the Gaza conflict. These systems were designed to support, not replace, human commanders.³⁸ However, their known error margins did not prevent them from accelerating target selection and engagement to as little as 20 seconds per decision.³⁹ The inadequate time for evaluation and deliberation meant that decisions to use force were often expedited (or ‘green-lit’), even when information was incomplete or ambiguous. In practice, AI-generated outputs were treated not as recommendations but as tacit approvals, granting AI decision-making authority akin to a commander’s order, without valid moral justification or ethical accountability.

While this article focuses primarily on automation bias and its psychological effects on human decision-makers, other forms of bias—most notably algorithmic bias—are equally consequential. Algorithmic bias refers to systemic distortions that arise from biased training data, flawed model design or socio-technical feedback loops.⁴⁰ These biases can lead to ethically unjust outcomes, particularly in areas such as target selection or threat prioritization where opaque or flawed reasoning may go unchallenged. Addressing these risks requires not only technical safeguards but also robust ethical frameworks, transparency mechanisms and sustained institutional oversight. The hybrid ethical model introduced below is intended not only to support human moral judgement but also to inform the design of AI-DSS systems capable of recognizing and mitigating these deeper structural biases. Although no ethical framework can eliminate algorithmic bias (or the other biases discussed), the proposed hybrid model seeks to mitigate its impact by embedding principles of fairness,

³⁵ David Lyell and Enrico Coiera, ‘Automation bias and verification complexity: a systematic review’, *Journal of the American Medical Informatics Association* 24: 2, 2017, pp. 423–31, <https://doi.org/10.1093/jamia/ocw105>.

³⁶ Kevin Baum, Susanne Mantel, Eva Schmidt and Timo Speith, ‘From responsibility to reason-giving explainable artificial intelligence’, *Philosophy & Technology* 35: 12, 2022, <https://doi.org/10.1007/s13347-022-00510-w>.

³⁷ Tianling Xie and Pentina Iryna, ‘Attachment theory as a framework to understand relationships with social chatbots: a case study of Replika’, in *Proceedings of the 55th Hawaii International Conference on System Sciences*, 2022, pp. 2046–55, <https://doi.org/10.24251/HICSS.2022.258>.

³⁸ Dorsey and Bo, ‘AI-enabled decision-support systems in the joint targeting cycle’.

³⁹ Harry Davies, Bethan McKernan and Dan Sabbagh, ‘“The Gospel”: how Israel uses AI to select bombing targets in Gaza’, *Guardian*, 1 Dec. 2023, <https://www.theguardian.com/world/2023/dec/01/the-gospel-how-israel-uses-ai-to-select-bombing-targets>.

⁴⁰ See Alexander Blanchard and Laura Bruun, *Bias in military artificial intelligence* (Stockholm: Stockholm International Peace Research Institute, 2024), <https://www.sipri.org/publications/2024/sipri-background-papers/bias-military-artificial-intelligence>; Ingvild Bode, ‘Falling under the radar: the problem of algorithmic bias and military applications of AI’, *Humanitarian Law & Policy* blog, 14 March 2024, <https://blogs.icrc.org/law-and-policy/2024/03/14/falling-under-the-radar-the-problem-of-algorithmic-bias-and-military-applications-of-ai>; Bode and Bhila, ‘The problem of algorithmic bias in AI-based military decision support systems’.

accountability and transparency into AI-DSS design, while promoting continuous human oversight and critical evaluation of system outputs.

Several experts argue that if humans adopt a passive supervisory role in AI-assisted decision-making, the distinction between AI-DSS and autonomous weapon systems—which select and apply force without human intervention—becomes increasingly blurred.⁴¹ This highlights a fundamental tension: while AI-DSS is designed to enhance military decision-making by improving speed and computational precision, meaningful ethical deliberation requires time for careful reflection. This contradiction is especially problematic given AI's current technical limitations. An over-reliance on AI-DSS risks undermining compliance with IHL, potentially increasing civilian casualties and eroding human moral agency in critical military decisions.

AI-assisted decision-making may also reshape rules of engagement, potentially lowering ethical thresholds for acceptable civilian harm.⁴² For example, Israel's Dahiya doctrine explicitly allows for disproportionate force, leading to extensive civilian casualties and the destruction of infrastructure.⁴³ As the Gaza conflict progressed, the IDF's threshold for 'acceptable' civilian deaths reportedly increased, highlighting a dangerous precedent: no universal ethical framework currently exists that can be fully integrated into AI-DSS to govern human-machine interactions while preserving human moral agency. Without robust oversight mechanisms (proposed below), AI-DSS risks widening the 'accountability gap', making responsibility for life-and-death decisions increasingly ambiguous.

Humans naturally anthropomorphize AI, attributing human-like qualities such as reasoning, morality and judgement to machines, even when such traits do not exist.⁴⁴ This misconception is reinforced by AI systems designed to mimic human behaviour, fostering misplaced trust in AI as an independent moral decision-maker.⁴⁵ For instance, a recent analysis of Israel's use of the Lavender AI-DSS tool in target selection described AI's actions as being carried out 'coldly by the machine'—as if AI was acting with its agency.⁴⁶ Such perceptions risk shifting AI-DSS from a decision-support tool to a de facto source of moral authority. The more AI is perceived as trustworthy and morally neutral,

⁴¹ See Marta Bo and Jessica Dorsey, 'Symposium on military AI and the law of armed conflict: the "need" for speed—the cost of unregulated AI decision-support systems to civilians', *Opinio Juris*, 4 April 2024; Peter Asaro, 'On banning autonomous weapon systems: human rights, automation, and the dehumanization of lethal decision-making', *International Review of the Red Cross* 94: 886, 2012, <https://doi.org/10.1017/S1816383112000768>; Boulanin, Brockmann and Richards, *Responsible artificial intelligence research and innovation*.

⁴² Bo and Dorsey, 'Symposium on military AI and the law of armed conflict'.

⁴³ Ayşe Işın Kirenci, "'Cheap stupid bombs': what's the Dahiya doctrine in Israel's war on Gaza?", *TRT World*, 21 Dec. 2023, <https://www.trtworld.com/article/16332266>.

⁴⁴ Fred Kiel and Doug Lennick, *Moral intelligence: enhancing business performance and leadership success* (Hoboken, NJ: Prentice Hall, 2007, first publ. 2005).

⁴⁵ James Johnson, 'Is there a human in the machine? AI and future warfare', *War on the Rocks*, 30 May 2024, <https://warontherocks.com/2024/05/is-there-a-human-in-the-machine-ai-and-future-warfare>.

⁴⁶ Bethan McKernan and Harry Davies, "'The machine did it coldly': Israel used AI to identify 37,000 Hamas targets", *Guardian*, 3 April 2024, <https://www.theguardian.com/world/2024/apr/03/israel-gaza-ai-database-hamas-airstrikes>.

the more likely it is that users will uncritically accept its recommendations, even when those recommendations raise ethical concerns.

As Robert Sparrow and Adam Henschke argue, overtrusting AI in decision-making could result in AI-DSS tools being treated as if they ‘will come to have the psychological, or even normative and institutional, force of orders’—thus blurring the boundary between AI-generated advice and legitimate military command.⁴⁷ The following section introduces a framework for addressing these socio-technical challenges and outlines a method for applying various ethical approaches to regional crisis scenarios that will be explored later in the article.

Humans in the machine: an ethical framework for human-machine interactions

As outlined earlier, the key challenge is to harness AI’s computational capabilities in crisis decision-making while preserving human moral agency. The goal is not to replace human decision-makers, but rather to create a collaborative interface where AI can support and enhance human ethical reasoning without undermining fundamental moral responsibility. This design challenge carries a second-order responsibility: ensuring accountability for individual commanders and those *who act with and through* AI-DSS. This section does not primarily address the philosophical debate over whether AI can be constructed as an artificial moral agent capable of ethical reasoning and autonomous decision-making.⁴⁸ Instead, it focuses on the epistemological and ontological challenges of developing AI-DSS tools that can perform context-sensitive ethical reasoning—what James Moor terms ‘functional ethics’—to augment, rather than replace, human decision-making in military crises.⁴⁹ As discussed earlier, these twin goals are not mutually exclusive, but intrinsic to the socio-technical dynamics of human-machine interactions. Successfully balancing them is critical to ensuring military commanders retain meaningful moral agency in future warfare.

Moor’s hierarchy of ethical agents

James Moor’s four-tier schema offers a helpful framework for categorizing ethical agency in machines.⁵⁰ His hierarchy distinguishes between ethical impact agents, implicit, explicit and full ethical agents.⁵¹ Ethical impact agents refer to any machine that performs a task and produces effects that can be ethically evaluated, even if the machine has no ethical awareness. Implicit ethical agents, by contrast, act in ways that can be considered ethical or unethical due

⁴⁷ Sparrow and Henschke, ‘Minotaurs, not centaurs’, p. 123.

⁴⁸ Wendell Wallach, Colin Allen and Iva Smit, ‘Machine morality: bottom-up and top-down approaches for modelling human moral faculties’, *AI and Society* 22: 4, 2008, pp. 565–82, <https://doi.org/10.1007/s00146-007-0099-0>.

⁴⁹ J. H. Moor, ‘The nature, importance, and difficulty of machine ethics’, *IEEE Intelligent Systems* 21: 4, 2006, pp. 18–21, <https://doi.org/10.1109/MIS.2006.80>

⁵⁰ Moor, ‘The nature, importance, and difficulty of machine ethics’, pp. 18–21.

⁵¹ Moor, ‘The nature, importance, and difficulty of machine ethics’, p. 19.

to their programming or inherent capabilities, making them the minimum ethical standard for machines interacting with humans. In the military domain, Ronald Arkin's concept of an 'ethical governor' for autonomous military robots applies Moor's implicit ethical agent model. However, it has only been tested in constrained simulated environments.⁵²

Explicit ethical agents, the next level in Moor's framework, can 'represent ethics explicitly and then operate effectively based on this knowledge'.⁵³ This would require AI systems to be programmed with ethical reasoning frameworks, such as 'deontic logic', which concerns what is permitted, obligatory or forbidden in moral decision-making.⁵⁴ While Moor and other scholars argue that explicit ethical agency should be a long-term goal of machine ethics, achieving this level of ethical processing remains a significant technical and philosophical challenge.⁵⁵ The final category, full ethical agents, would in the most futuristic scenario be able to reason ethically, justify moral decisions and act as autonomous moral beings. However, many scholars argue that machines, lacking consciousness, intentionality and free will, cannot be full ethical agents like humans.⁵⁶ Even if machines could perform ethically competent behaviour, their lack of human-like moral understanding, reflection and affective sensitivity makes independent machine moral agency conceptually and practically problematic.⁵⁷

Given the rapid pace of AI development—especially with the emergence of LLMs—there is an increasingly urgent need to design ethical frameworks for both implicit and explicit ethical agents. AI-DSS should be capable of operationalizing ethical principles in a manner that provides meaningful information for high-stakes decision-making. However, these systems cannot autonomously resolve moral dilemmas, nor can they be held morally responsible for their outputs. The moral burden of decisions involving the use of force must remain with human agents. Like advances in AI more broadly, progress up Moor's hierarchy of machine ethics will likely be incremental, uneven and unpredictable, making it all the more critical to confront the ethical implications of today's design choices before they become deeply embedded in future systems. This shift raises a central design imperative: how can we build AI systems that support ethical reasoning without displacing human moral agency? Ensuring

⁵² Arkin's 'the ethical governor' is a system architecture that is designed to simulate robots making decisions on the use of force by reference to a set of programmed ethical constraints and internal controls in a manner consistent with the laws of war and rules of engagement: see Ronald C. Arkin, *Governing lethal behavior: embedding ethics in a hybrid deliberative/reactive robot architecture* (Atlanta: Georgia Institute of Technology, 2009).

⁵³ Moor, 'The nature, importance, and difficulty of machine ethics', p. 20.

⁵⁴ Research has been conducted on programming deontic logic into algorithms for wargaming simulations. Konstantine Arkoudas, Selmer Bringsjord and Paul Bello, 'Toward ethical robots via mechanized deontic logic', in *Proceedings from the 2005 AAAI Fall Symposium* (Menlo Park, CA: AAAI Press, 2006).

⁵⁵ Peter M. Asaro, 'What should we want from a robot ethic?', *International Review of Information Ethics*, vol. 6, 2006, pp. 9–16, <https://doi.org/10.29173/iriet134>; Wallach and Allen, *Moral machines*.

⁵⁶ Andreas Matthias, 'Algorithmic moral control of war robots: philosophical questions', *Law, Innovation and Technology* 3: 2, 2011, pp. 279–301, <https://doi.org/10.5235/175799611798204923>.

⁵⁷ Martin Gibert, 'The case for virtuous robots', *AI Ethics*, vol. 3, 2023, p. 135–44 at p. 136, <https://doi.org/10.1007/s43681-022-00185-1>.

that commanders retain meaningful control over AI-assisted decision-making is essential to prevent the unintended outsourcing of moral responsibility—particularly if the current human-in-the-loop consensus holds in future applications of force. Yet this also brings to the surface a deeper ethical tension: how can we prevent the erosion of human moral accountability, when aspects of ethical reasoning are *partially delegated* to machines?

Top-down vs bottom-up

Wendell Wallach and Colin Allen distinguish between two primary approaches to programming ethics into AI systems: top-down (rule-based) and bottom-up (learning-based) models.⁵⁸ Top-down approaches apply predefined ethical rules from deontology, consequentialism or virtue ethics to guide AI decision-making.⁵⁹ These models function as heuristics that translate abstract ethical principles into operational guidelines.⁶⁰ However, researchers continue to debate whether ethical codes, norms and standards can be effectively encoded into AI in ways that ensure consistency, flexibility and context-sensitivity.⁶¹

Bottom-up approaches, by contrast, use machine learning and reinforcement learning to develop ethical behaviour through experience rather than relying on fixed principles. Like human learning, these models refine their decision-making processes through trial and error, receiving feedback based on the moral consequences of their actions.⁶² While this allows for adaptability, it also introduces unpredictability and raises concerns about alignment with human values.⁶³ In practice, most AI-DSS employ a hybrid approach, combining structured ethical guidelines with adaptive learning mechanisms. This integration aims to strike a balance between the predictability of rule-based ethics and the flexibility of machine learning. LLMs, such as ChatGPT, primarily use a top-down structure but incorporate bottom-up elements by identifying patterns and refining responses based on human interaction and training data.

⁵⁸ Wallach and Allen, *Moral machines*, pp. 79–81.

⁵⁹ An additional prominent deontological approach in the machine ethics literature is Asimov's 'laws of robotics': see Isaac Asimov, *I, robot* (New York: Gnome Press, 1950).

⁶⁰ Allen Newell and Herbert A. Simon, 'Computer science as empirical inquiry: symbols and search', *Communications of the ACM* 19: 3, 1976, pp. 113–26, <https://doi.org/10.1145/360018.360022>.

⁶¹ Amanda Sharkey, 'Can we program or train robots to be good?', *Ethics & Information Technology*, vol. 22, 2020, pp. 283–95, <https://doi.org/10.1007/s10676-017-9425-5>.

⁶² Sharkey, 'Can we program or train robots to be good?'.

⁶³ A recent study by OpenAI found that certain internal features in AI models act like 'personas' linked to unsafe behaviour (e.g. toxicity or deception). By either amplifying or suppressing these features, the researchers could causally influence the model's alignment. Importantly, alignment could be restored with just a few hundred fine-tuning examples ('emergent re-alignment'), suggesting a viable interpretability-driven strategy for early detection and correction of misalignment during training: see 'Toward understanding and preventing misalignment generalization', OpenAI, 18 June 2025, <https://openai.com/index/emergent-misalignment>.

Technical bottlenecks

Implementing ethical decision-making in AI-DSS presents significant technical challenges. One of the most pressing issues is the ability to predict and assess the consequences of actions in real-time combat scenarios.⁶⁴ Unlike humans, AI systems lack embodiment in the natural world, raising fundamental questions: how can an AI evaluate right from wrong? Who determines the ethical rules it must follow? These concerns are particularly relevant in high-pressure military situations, where moral considerations must be balanced against operational effectiveness.

A classic ethical dilemma that illustrates these challenges is a military variant of the trolley problem, in which an AI-DSS must decide whether to authorize a missile strike on a high-value target, knowing that civilian casualties are likely to occur.⁶⁵ The system would need to navigate competing ethical frameworks, including deontological principles that emphasize the inviolability of civilian life, consequentialist calculations aimed at preventing future harm and virtue-based considerations regarding military honour and the principle of proportionality. Unlike human decision-makers, who can intuitively integrate multiple ethical considerations, AI must rely on predefined weightings that may not fully capture the moral complexity of the situation.

To address these concerns, some scholars propose hybrid ethical models that blend deontological constraints with consequentialist evaluations, an approach sometimes referred to as ‘limited consequentialism’ within just war theory.⁶⁶ However, a fundamental challenge remains: how should AI-DSS determine when consequentialist calculations override deontological imperatives? Beyond computational difficulties, AI-based decision support must also account for human psychological and social factors, ensuring ethical decision-making aligns with military norms and broader social values. Moreover, for AI systems to earn trust and confidence, their guidance must align with social acceptability. As Wallach and Allen note, ‘people don’t want artificial moral agents to replicate the abstractions of moral philosophers. Instead, they want machines to respond flexibly and sensitively in real and virtual environments.’⁶⁷

The ‘framing problem’ further complicates AI ethics by highlighting AI’s difficulty in determining which ethical considerations are relevant in complex real-world situations.⁶⁸ Moral decision-making requires AI systems to assess the relevance of available information, predict the consequences of different actions and account for human cognitive and emotional factors.⁶⁹ Kwame Appiah

⁶⁴ Wallach and Allen, *Moral machines*, p. 96.

⁶⁵ The ‘trolley problem’ is a classic ethical thought experiment in which one must decide whether to divert a runaway trolley to a track where it will kill one person instead of five, posing a moral dilemma between actively causing harm to save more lives and passively allowing greater harm to occur.

⁶⁶ Just war theory also provides the philosophical foundation for IHL.

⁶⁷ Wallach and Allen, *Moral machines*, p. 215.

⁶⁸ Anthony F. Beavers, ‘Moral machines and the threat of ethical nihilism’, in Patrick Lin, Keith Abney and George Bekey, eds, *Robot ethics: the ethical and social implications of robotics* (Cambridge, MA: MIT Press, 2014, first publ. 2012), p. 335.

⁶⁹ Patrick Lin, Keith Abney and George Bekey, ‘Robotics, ethical theory, and metaethics: a guide for the

underscores this challenge, noting that ‘the act of framing ... of describing a situation, and thus of *determining that there’s a decision to be made*—is itself a moral task. It’s often *the* moral task.’⁷⁰

Scalability and adaptability remain significant challenges in the ethical implementation of AI-DSS. Ethical rules must be broad enough—for example, the principle of ‘do not harm civilians’—to apply across diverse scenarios while remaining flexible enough to adapt to evolving battlefield conditions. Overly rigid rule-based approaches risk making AI impractical for military operations, while excessive reliance on heuristics could weaken the specificity and rigour of ethical decision-making. Ensuring AI-DSS can acquire and update ethical knowledge over time is an ongoing challenge. While machine-learning techniques such as Bayesian reasoning, supervised learning and reinforcement learning offer potential solutions, they also introduce risks. AI systems may develop unintended or emergent behaviours that have not been explicitly tested or verified, leading to unpredictable ethical consequences. Incorporating heuristics—such as analysing facial expressions, voice patterns and social cues—into AI decision-making may help navigate ethical dilemmas, but it risks compromising the precision of ethical reasoning.⁷¹

Finally, acquiring and maintaining up-to-date background information presents a significant technical challenge. AI systems use machine-learning techniques (for example, Bayesian reasoning, supervised learning and reinforcement learning) to refine their ethical decision-making over time.⁷² However, the challenge of determining which preferences are ethically permissible remains unresolved. For instance, a military commander may prioritize operational success over minimizing collateral damage—should an AI system align with such preferences? While online learning, model fine-tuning and reinforcement learning offer potential solutions, experts warn that AI systems may develop emergent behaviours that have not been tested or verified, leading to alignment issues and unpredictable consequences.⁷³

Philosophical approaches

AI-DSS ethical frameworks are typically structured around three dominant moral theories: consequentialism, deontology and virtue ethics. Consequen-

perplexed’, in Lin, Abney and Bekey, *Robot ethics*, p. 45.

⁷⁰ Kwame Anthony Appiah, *Experiments in ethics* (Cambridge, MA: Harvard University Press, 2008), p. 196. (First emphasis added, second in original.)

⁷¹ Wendell Wallach, ‘Robot minds and human ethics: the need for a comprehensive model of moral decision making’, *Ethics and Information Technology* 12: 3, pp. 243–50 at p. 247, <https://doi.org/10.1007/s10676-010-9232-8>.

⁷² For example, Stuart Russell has proposed using a reinforcement learning technique to maximize user preferences: Stuart Russell, *Human compatible: artificial intelligence and the problem of control* (New York: Viking, 2019).

⁷³ For example, recent instances of ‘reward hacking’ reveal how AI systems can exploit scoring loopholes in tasks assigned by users to maximize rewards—knowingly deviating from user intent—highlighting the dangers of emergent misalignment even in systems capable of understanding their own deception: ‘Recent frontier models are reward hacking’, METR blog, 5 June 2025, <https://metr.org/blog/2025-06-05-recent-reward-hacking>.

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tialism, particularly in its utilitarian form, seeks to maximize overall benefits by evaluating the outcomes of decisions and actions.⁷⁴ However, this approach presents computational and ethical challenges in military applications. Predicting long-term consequences, often termed the ‘ripple effects’ of an action, is highly complex and frequently unreliable.⁷⁵ AI-driven utilitarian calculations therefore struggle to account for the unpredictability and unintended consequences of warfare, making them an unstable foundation for ethical decision-making. Even if these technical obstacles were resolved, utilitarianism in warfare raises deeper moral concerns. The distinction between actively killing civilians and allowing civilians to die is difficult to establish within utilitarian logic, assuming such a distinction even exists. Stephen Deakin argues that utilitarianism risks becoming ‘the ethic of a highly educated rational calculator’, an approach that reduces harm assessment to mathematical optimization constrained only by military objectives and operational feasibility.⁷⁶

While utilitarianism is not inherently flawed, its flexibility creates opportunities for exploitation, mainly when applied to military decision-making. For example, reports from the Israel– Hamas conflict suggested that AI-DSS was used to predefine acceptable levels of civilian casualties for targeting lower-ranking Hamas leaders. One article has claimed that these thresholds increased over time, allowing for up to 20 civilian deaths per targeted militant.⁷⁷ This raises serious concerns about how utilitarian logic can normalize escalating collateral damage, as proportionality calculations may gradually shift over time to accommodate tactical objectives.

Deontology, by contrast, relies on fixed moral principles to guide decision-making, such as Kant’s ‘categorical imperative’, which demands adherence to universal moral rules.⁷⁸ Deontology ensures explicit ethical constraints in military applications, such as the prohibition against intentionally targeting civilians—for example, an AI-DSS programmed to reject any strike plan involving foreseeable harm to non-combatants, regardless of strategic value. While this approach provides a strong ethical foundation, its rigidity can make it impractical in fluid combat environments where decision-making requires situational flexibility.⁷⁹

⁷⁴ Utilitarianism is a consequentialist ethical theory, most closely associated with John Stuart Mill and Jeremy Bentham, that claims the best action is the one that maximizes overall happiness or ‘utility’, often summarized as achieving ‘the greatest good for the greatest number’: Jeremy Bentham, *An introduction to the principles of morals and legislation* (London: T. Payne and Son, 1789).

⁷⁵ Michael Walzer, *Just and unjust wars: a moral argument with historical illustrations* (New York: Basic Books, 1977).

⁷⁶ Stephen Deakin, ‘Education in an ethos at the Royal Military Academy Sandhurst’, in Paul Robinson, Nigel de Lee and Don Carrick, eds, *Ethics education in the military* (Aldershot: Ashgate, 2008), p. 24.

⁷⁷ McKernan and Davies, “‘The machine did it coldly’”.

⁷⁸ Immanuel Kant’s ‘categorical imperative’ is a foundational concept in his moral philosophy, which asserts that behaviour must be guided by universal principles that can be applied consistently by everyone. Immanuel Kant, *Groundwork for the metaphysics of morals*, transl. by Mary Gregor (Cambridge, UK: Cambridge University Press, 1998, first publ. 1785).

⁷⁹ Early-stage research on programmable deontological rules is well advanced. Selmer Bringsjord and Joshua A. Taylor, ‘The divine-command approach to robot ethics’, in Lin, Abney and Bekey, *Robot ethics*, pp. 251–67.

Virtue ethics, in contrast to consequentialism and deontology, emphasizes moral character and the development of adaptive decision-making. Rather than prescribing fixed rules or maximizing predefined outcomes, virtue ethics focuses on cultivating ethical judgement sensitive to the complexities of specific situations. For instance, an AI-DSS guided by virtue ethics might weigh restraint, courage or prudence when advising a commander facing an ambiguous threat in a populated area. This approach is particularly well suited to military AI applications, where ethical dilemmas are often context-dependent and cannot be resolved through strict adherence to rules or algorithmic optimization. In practice, however, no single ethical framework is sufficient for AI-DSS. Instead, these systems require a hybrid approach that integrates deontological constraints, consequentialist risk assessments and virtue-based adaptability to balance military necessity with ethical oversight.

A hybrid approach

Virtue ethics bridges the gap between rule-based and outcome-driven approaches, providing a more flexible and context-sensitive framework for ethical decision-making. Unlike deontology, which prioritizes adherence to strict moral duties, or consequentialism, which focuses on maximizing positive outcomes, virtue ethics emphasizes cultivating moral character and practical wisdom. Aristotle's concept of the 'golden mean' illustrates this approach, advocating for ethical decision-making that avoids rigid absolutes while maintaining ethical integrity.⁸⁰ While deontology, consequentialism and virtue ethics represent distinct philosophical traditions, this article follows a pluralist, pragmatic approach, recognizing that no single theory can address the full range of ethical challenges in crisis decision-making. The proposed hybrid exemplar framework leverages the strengths of each tradition to inform the design of an ethical AI-DSS, without compromising their theoretical distinctions.

Michael Slote characterizes a virtuous person not as someone who follows ethical rules, but as someone who demonstrates intelligence and moral sensitivity in determining the right course of action in different circumstances.⁸¹ This adaptability makes virtue ethics particularly relevant to human–AI interactions in military decision-making. While rule-based and utilitarian models provide structured ethical guidance, they struggle to accommodate the complexities of dynamic military environments, moral ambiguity and shifting contextual factors.⁸²

Recent scholarship has explored ways to integrate virtue ethics into AI systems. William Bauer, for instance, has proposed a model of 'two-level utili-

⁸⁰ In the Aristotelian tradition, moral agents develop virtuous behaviour not primarily through explicit propositional teaching but through habituation from early childhood.

⁸¹ Michael Slote, 'Virtue ethics', in: Marcia W. Baron, Philip Pettit and Michael Slote, *Three methods of ethics: a debate* (Oxford: Blackwell, 1997), p. 178.

⁸² Bongani Andy Mabaso, 'Artificial moral agents within an ethos of AI4SG', *Philosophy and Technology*, vol. 34, 2021, pp. 7–21, <https://doi.org/10.1007/s13347-020-00400-z>.

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tarianism', which combines virtue-based reasoning with utility calculations to guide AI decision-making.⁸³ However, designing AI systems that align with human virtues requires more than simply encoding a single ethical framework into an algorithm. Iason Gabriel argues that the goal should not be to impose a universal moral system onto AI, but to develop systems capable of navigating ethical diversity.⁸⁴ Different cultures and individuals hold reasonable yet differing beliefs about values. AI-DSS must be designed to recognize and integrate these variations, rather than being locked into rigid ethical models.

Algorithmically implementing virtue ethics does not require AI to possess human-like moral character. Instead, it involves designing AI-DSS that exhibit behaviours consistent with ethical judgement through structured learning processes. To achieve this, researchers increasingly advocate for bottom-up machine-learning approaches, such as reinforcement learning and supervised learning, instead of purely top-down rule-based systems.⁸⁵ Instead of relying on pre-programmed ethical principles, AI-DSS can analyse and learn from patterns in human ethical decision-making, historical case-studies and ethical discourse across disciplines. This approach allows AI to develop ethical responsiveness dynamically, rather than being restricted to static moral codes that may fail in complex, unpredictable situations.

For AI systems to approximate virtuous behaviour, they must move beyond abstract moral theorizing and engage with real-world ethical dilemmas. Virtues such as courage, prudence, discipline, mercy and restraint can be modelled within rule-based frameworks. Still, true adaptability requires machine-learning mechanisms that allow AI to interpret and apply ethical principles across diverse contexts. One approach to this challenge is demonstrated by MedEthEx, a machine-learning system designed to assist healthcare professionals in resolving ethical dilemmas.⁸⁶ Similarly, Bertram Malle has proposed that AI can develop moral competency by analysing ethical reasoning in literature, films, interviews and real-world human decision-making processes.⁸⁷

To address these challenges, Andy Mabaso proposes 'ethical exemplarism', a method that enables AI to learn from real-world examples of ethical behaviour rather than relying on rigid moral rules. This approach eschews a fixed moral code in favour of three ethical dimensions: goodness, drawn from virtue ethics; rightness, aligned with deontological principles; and utility, associated with consequentialist reasoning.⁸⁸ By combining these elements, ethical exemplars

⁸³ William A. Bauer, 'Virtuous vs. utilitarian artificial moral agents', *AI & Society*, vol. 35, 2020, pp. 263–71, <https://doi.org/10.1007/s00146-018-0871-3>.

⁸⁴ Iason Gabriel, 'Artificial intelligence, values, and alignment', *Minds & Machine*, vol. 30, 2020, pp. 411–37, <https://doi.org/10.1007/s11023-020-09539-2>.

⁸⁵ Gibert, 'The case for virtuous robots', p. 136.

⁸⁶ Michael Anderson, Susan Leigh Anderson and Chris Armen, 'MedEthEx: a prototype medical ethics advisor', Proceedings of the Eighteenth Conference on Innovative Applications of Artificial Intelligence, Aug. 2006, <https://aaai.org/papers/01759-aaaio6-292-medethex-a-prototype-medical-ethics-advisor>.

⁸⁷ Bertram F. Malle, 'Integrating robot ethics and machine morality: the study and design of moral competence in robots', *Ethics & Information Technology*, vol. 18, 2016, <https://doi.org/10.1007/s10676-015-9367-8>.

⁸⁸ Mabaso, 'Artificial moral agents within an ethos of AI4SG'.

provide a computationally feasible framework for machine moral reasoning that is both ethically pluralistic and socially intelligible. Unlike conventional rule-based AI systems, ethical exemplars enable situational flexibility, allowing AI to shift between ethical frameworks based on the specific operational context.⁸⁹ For instance, deontological constraints might take precedence in scenarios requiring strict compliance with international humanitarian law, while in other cases, consequentialist reasoning might guide risk assessment and tactical decision-making. As Mabaso argues, ‘from a purely computational perspective, learning from example is more feasible than programming abstract ethical concepts’.⁹⁰

Building on this theoretical foundation, the article adapts ethical exemplarism to the domain of military AI decision support. The contribution lies in operationalizing this triadic ethical structure—goodness, rightness and utility—within AI-DSS interfaces that support commanders during crisis decision-making. The article proposes that AI-DSS could draw selectively from each framework depending on the operational context: for example, deontological constraints might govern decisions involving international humanitarian law, while consequentialist logic may guide proportionality assessments and virtue-based reasoning could inform judgements under moral uncertainty. This context-sensitive implementation of exemplarism potentially offers a more flexible and ethically robust foundation for preserving human moral agency in complex, high-stakes military environments.

From a philosophical perspective, no single ethical framework enjoys universal acceptance. Utilitarians continue to debate how to quantify and compare utility across diverse contexts; deontologists disagree over which moral duties should be prioritized when principles conflict; and virtue ethicists hold differing views on which character traits constitute moral excellence. These enduring disagreements highlight the challenge of grounding AI ethics in a single tradition. Given these tensions—and the current limits of AI cognition—the ethical exemplar approach does not aim to create fully autonomous moral agents capable of human-like ethical reasoning. This objective remains both conceptually and computationally infeasible. Instead, it offers a structured, pluralistic method for AI systems to approximate ethical reasoning by drawing on real-world examples across moral traditions, while ensuring that ultimate moral responsibility remains with human decision-makers. By incorporating virtue-based learning and contextual flexibility, AI-DSS can better navigate morally complex scenarios without being constrained by the rigidity or incompleteness of any single ethical doctrine. This hybridization of ethical theories within a hybrid exemplar framework—explicitly tailored to the demands of military decision support—constitutes a novel contribution of this study, offering a pragmatic and ethically pluralist model for preserving human moral agency in high-stakes environments.

⁸⁹ Mabaso, ‘Artificial moral agents within an ethos of AI4SG’, p. 16–17.

⁹⁰ Mabaso, ‘Artificial moral agents within an ethos of AI4SG’, p. 12.

Operationalizing ethical safeguards in AI-DSS

Table 1 outlines key metrics and technical features designed to enhance human moral agency in AI-assisted military decision-making. This section provides a brief overview of these recommendations and their practical implications. Decision-making accountability can be strengthened using transparent decision logs, justification templates and real-time tracking tools. Automatically recording when and why human commanders accept or override AI recommendations creates an audit trail that reinforces human authorship and enables meaningful post hoc review. Such mechanisms help ensure that responsibility for decisions remains with the human operator rather than being implicitly transferred to the system.

Trust calibration is equally critical. Confidence scores, uncertainty indicators and justification fields help balance trust and oversight, reducing automation bias and indiscriminate rejection. For instance, when an AI-DSS flags a high-risk recommendation with only moderate confidence, it should prompt human deliberation or further corroboration. To support transparency and understandability, interactive explanation tools and layered detail displays can help users grasp not only *what* the system recommends but *why*. Integrated workflows that prioritize ethically salient information enable operators to engage in ethical reflection, even under pressure of time.

Effective ethical override mechanisms must extend beyond formal availability to enable practical and routine use. AI-DSS should incorporate clear feedback loops that allow users to understand how their input influences subsequent system behaviour. Features designed to mitigate cognitive biases—such as counterfactual prompts or alternative course generation—can further help challenge automation bias and anthropomorphism, as discussed earlier. Moreover, collaborative feedback loops, including team dashboards and iterative updates, promote shared ethical judgement, ensuring that responsibility is distributed across human–AI teams rather than being siloed in either domain. Recognizing the psychological impact on users—including moral stress, cognitive burden and emotional fatigue—can inform the development of stress-reducing interfaces and operator support tools.

Finally, in multi-agent scenarios, where multiple human and AI actors interact, ethical adaptability becomes essential. Context-sensitive algorithms capable of real-time adjustment help ensure that ethical safeguards remain effective as operational complexity increases. Taken together, these recommendations translate the proposed hybrid ethical framework into concrete design and operational principles, offering practical pathways for ensuring that AI-DSS strengthens, rather than undermines, human moral agency in military crisis decision-making.

Table 1: Metrics and technical features for enhancing human moral agency in human–AI DSS interaction

<i>Metric</i>	<i>Purpose</i>	<i>AI-DSS technical features</i>	<i>Benchmark/ threshold</i>
<i>Decision-making accountability</i>	Ensures human operators retain authorship and responsibility for decisions and outcomes	Transparent logs, justification templates, real-time AI tracking, time-stamped actions, and override ratios	
<i>Trust in DSS calibration</i>	Assesses whether humans exhibit appropriate trust in AI outputs, avoiding overreliance	Confidence and uncertainty indicators, explainability features, mandatory override justifications, and user trust metrics	Real-time decision tracking. E.g. an override rate >25% indicates healthy human engagement
<i>Transparency and understandability</i>	Assesses whether the system presents its reasoning and confidence clearly enough to support timely, ethically informed human decisions under pressure	Decision paths, interactive explanations with layered detail, and tracking AI confidence–human acceptance correlations	Confidence–scoring system (0–100%)
<i>Time for deliberation</i>	Evaluates how effectively the system balances moral reflection with time-sensitive decision demands	Integrated workflows, streamlined interfaces, and prioritized alerts for timely interaction	
<i>Ethical override frequency</i>	Measures how often humans override AI recommendations	Override mechanisms with clear feedback loops and justifications for AI recommendations	Override pattern analysis e.g. 70–90% acceptance rate for high confidence
<i>Identify and mitigate human cognitive bias^a</i>	Evaluates the system’s ability to reduce human biases such as automation bias and anthropomorphism	Counterfactual scenarios, alternative options, training, bias alerts	
<i>Collaborative feedback loops</i>	Monitors the quality and frequency of human–AI interaction in decision refinement	Feedback prompts, team dashboards, and adaptive, user-informed updates	
<i>Psychological impact on users</i>	Tracks users’ moral stress, emotional well-being, and satisfaction during interaction	Surveys, behaviour monitoring, and stress-reducing interface design	
<i>Ethical adaptability in multi-agent scenarios</i>	Tests the system’s ability to dynamically adjust to context-specific dilemmas or feedback	Real-time updates, contextual learning, and situation-specific adaptation	

^a Bias in decision-making may arise from the decision-maker, the surrounding context or a misalignment between the two. In both experimental settings and operational environments, decisions are frequently made under conditions of novelty, ambiguity and competing objectives. See Baruch Fischhoff, *Prejudices about bias*, working paper no. 85–3 (Eugene, OR: Decision Research, 1985), <http://doi.org/10.2139/ssrn.1560126>.

Conclusion

The integration of AI-DSS represents a transformative shift in military crisis decision-making. While these systems enhance decision-making through computational speed, pattern recognition and data integration, they also introduce complex socio-technical dynamics that risk eroding human moral agency and accountability, fundamentally challenging traditional notions of human control. The key challenge lies in balancing the efficiency of AI-assisted decision-making with the imperative of ethical oversight, ensuring that human moral judgement remains central in decisions involving the use of force.

This article advances the field of military AI ethics in three significant ways. First, it advances a hybrid exemplar ethical framework that integrates elements of virtue ethics, deontology and consequentialism, specifically adapted to meet the demands of military crisis decision-making. This pluralist, context-sensitive approach presents a novel pathway for safeguarding human moral agency amid the rising autonomy and complexity of AI. Second, it articulates normative design criteria for evaluating and preserving human moral agency in AI-assisted military operations, including decision accountability, trust calibration and the quality of human-machine interaction. Rather than attempting to quantify moral agency in a reductive sense, the framework offers practical benchmarks for assessing how AI-DSS design and deployment shape human ethical responsibility under operational pressure. Third, the article moves beyond theoretical analysis by offering concrete institutional and technical safeguards—such as mandatory ethical override mechanisms, dynamic trust calibration systems and specialized training programmes—to strengthen human oversight, counter cognitive vulnerabilities and ensure AI-DSS serve as instruments of ethical support.

As AI-DSS become increasingly embedded in military operations, military organizations must deepen their understanding of the evolving relationship between human decision-makers and AI systems. Rather than treating AI-DSS as passive tools or technical aids, they should be understood as shared cognitive entities—shaping and shaped by the human actors they support. Future research should prioritize empirically testing these ethical frameworks through controlled wargaming exercises, field studies of human-machine interaction in operational contexts and cross-cultural analyses of AI-DSS deployment. These efforts are essential to ensure that AI enhances, rather than erodes, human ethical judgement and accountability in the automated battlespace of the future.